Predicting Diabetes Readmissions Before They Happen

A clinical decision support proposal for reducing preventable 30-day readmissions among diabetes patients using evidence-based machine learning

Spring 2026: Paul Q (550) & Briggs P (550 & 551)

Assumptions & Facts

01

NYS Trauma 1 Hospital

State specific jurisdiction is very important. Not a Fed hospital.

02

EHR is EPIC

Most widely used EHR. EPIC is well integrated with AHDS (Azure)

03

The Audience

Mix of hospital's clinical and technical leadership

04

Baseline data is from 1999-2008

Training and creating a ML model on it for 2026 > gaps

05

Pre-pilot

The ML model we built is pre pilot.

06

Pilot proposal

This is a proposal to launch a pilot ML project > fills data gaps

07

Goals not yet accomplished

Achievable goals will be identified

08

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Next ~45 minutes

01 **The Clinical Problem**
Why current processes miss high-risk patients

02 **Our Solution**
A proven ML model embedded directly in Epic EHR

03 **The Evidence**
Model performance translated to patient outcomes

04 **How It Works**
The model development journey - data to deployment

05 **Workflow Integration**
What clinicians will see & when

06 **Architecture & Security**
7-phase implementation with NYS compliance

07 **Governance & Safety**
Human-in-the-loop oversight & change management

08 **Next Steps**
Proposal, timeline & investment ask

Credible literature on this subject

01

BMC Medical Informatics & Decision Making

2021 - RF model best predicts readmissions (top AUC)

02

Computer Modeling in Engineering & Sciences

2025 - RF best: 96% readmission accuracy & 87% LOS prediction.

03

Cureus Journal of Medical Sciences

2025 - XGBoost boosts ML prediction of 30-day readmissions using structured EHR.

04

Information Systems Engineering & Management

2025 - Ensemble model achieved the highest accuracy (98.6%)

05

Semantic Scholar

Database with multiple papers on ML models on this subject .

06

Hugging Face

2024 - Cutting edge ML model for <30 day readmission prediction

07

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08

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01 The Clinical Problem

20%

of diabetes patients
readmitted within 30 days [1]

$15K+

average cost per
preventable readmission [2]

$2M+

in annual CMS penalties
for NYS hospitals - above benchmark [3]

01 The Clinical Problem

The Core Problem

Lack of a systematic, data-driven way to identify which diabetic patients are at highest risk of prolonged readmission, before discharge. Decisions rely on clinical intuition alone, leading to **missed at-risk patients, resource strain, avoidable readmissions, and $'s.**

Current discharge planning is reactive. High-risk patients are identified only after they return to the ED, if at all.

An AI risk flag at the point of discharge changes that.

50%

of readmissions may be preventable with early intervention [4]

02 Our Solution

A Risk Flag in the Clinical Workflow

In the clinical lifecycle, including when the physician initiates discharge, the tool silently scores that patient's 30-day readmission risk using data already in Epic.

If the risk is high (40%), a plain-language alert appears in the clinical/discharge workflow with a one-click care plan.

No new login

Launches inside Epic

Plain language

No algorithm jargon

Advisory only

Physician decides



02 Our Solution

A Clinical Decision Support Tool - Not Just an Algorithm

The model flags high-risk patients at discharge so care teams can intervene before a readmission occurs.



Why this matters:

- The clinician remains in full control - the model provides a second opinion, not a final decision.
- High-risk alerts prompt targeted follow-up care before the patient ever leaves the hospital.

02 Our Solution - Data Foundation & Feature Design

The Data Foundation

Dataset	UCI 130-US Hospitals Diabetes
Records	101,766 patient encounters
Time Period	1999-2008 (10 years)
Variables	50 clinical & demographic features
Target	Readmission <30 days
Quality	Cleaned, deduplicated, mapped, imputed, encoded, bias-checked (balanced)

Clinically Meaningful Features

Prior Utilization

Number of inpatient & ER visits, prior admissions

Medication Profile

Insulin use, # medications changed, diabetes drugs

Comorbidities

Diagnoses count, primary/secondary/tertiary Diag w/ ICD codes

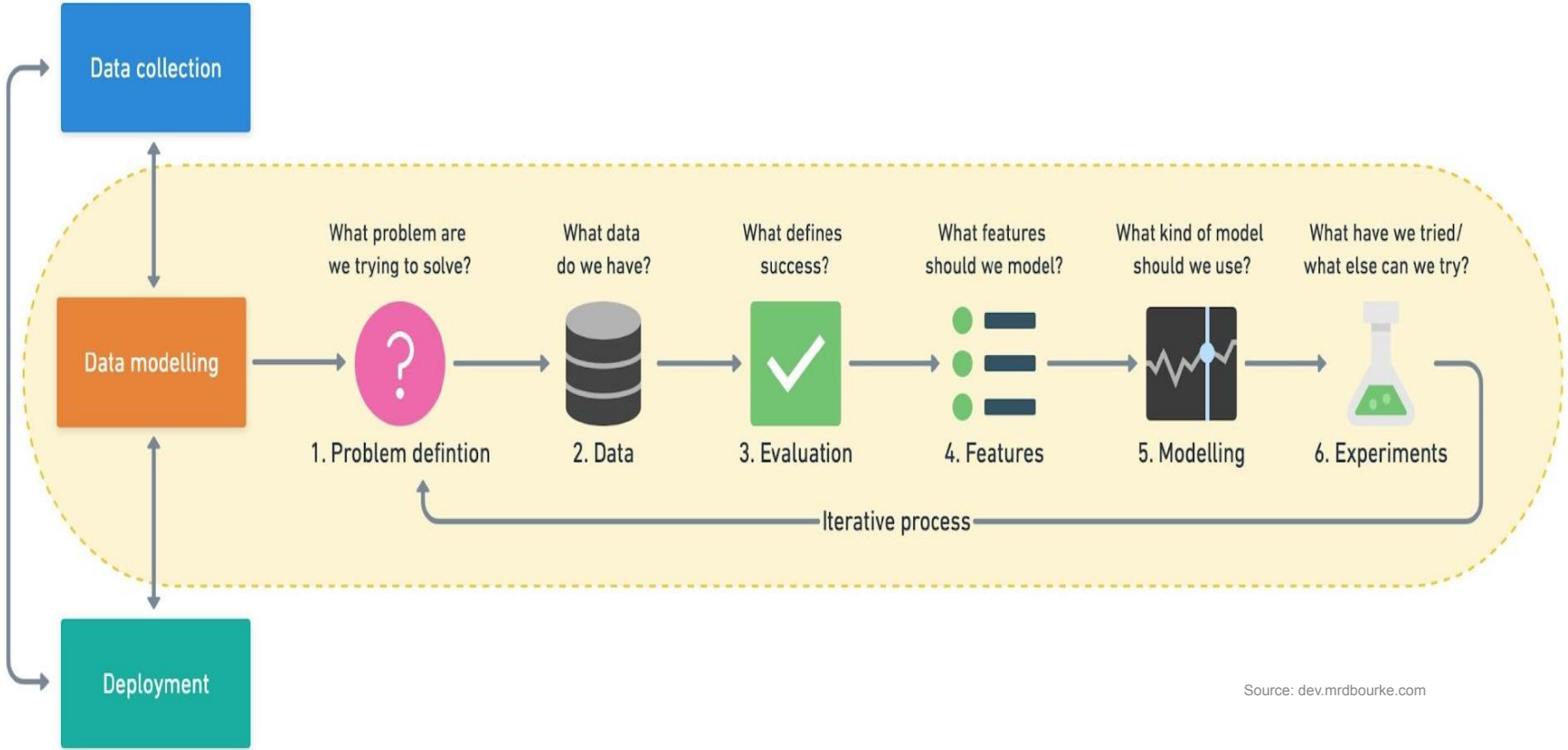
Care Intensity

Hospital stay duration, lab procedures ordered

Demographics

Age group, payer mix, admission/discharge source

02 The ML Framework: From Data to Deployment



Source: dev.mrdbourke.com

02 How the model was built

Four rigorous phases ensure the model is reliable, unbiased, and ready for clinical use.

1

Explore & Clean

- ✓ 101K+ records reviewed
- ✓ Missing values imputed
- ✓ Duplicates/empties removed
- ✓ Outliers handled

2

Feature Engineering

- ✓ 23 meaningful features
- ✓ Clinical groupings created
- ✓ Encoding applied
- ✓ Class imbalance corrected

3

Model Training

- ✓ 9 algorithm types tested
- ✓ Includes Ensemble models
- ✓ Cross-validation
- ✓ Fair comparison baseline

4

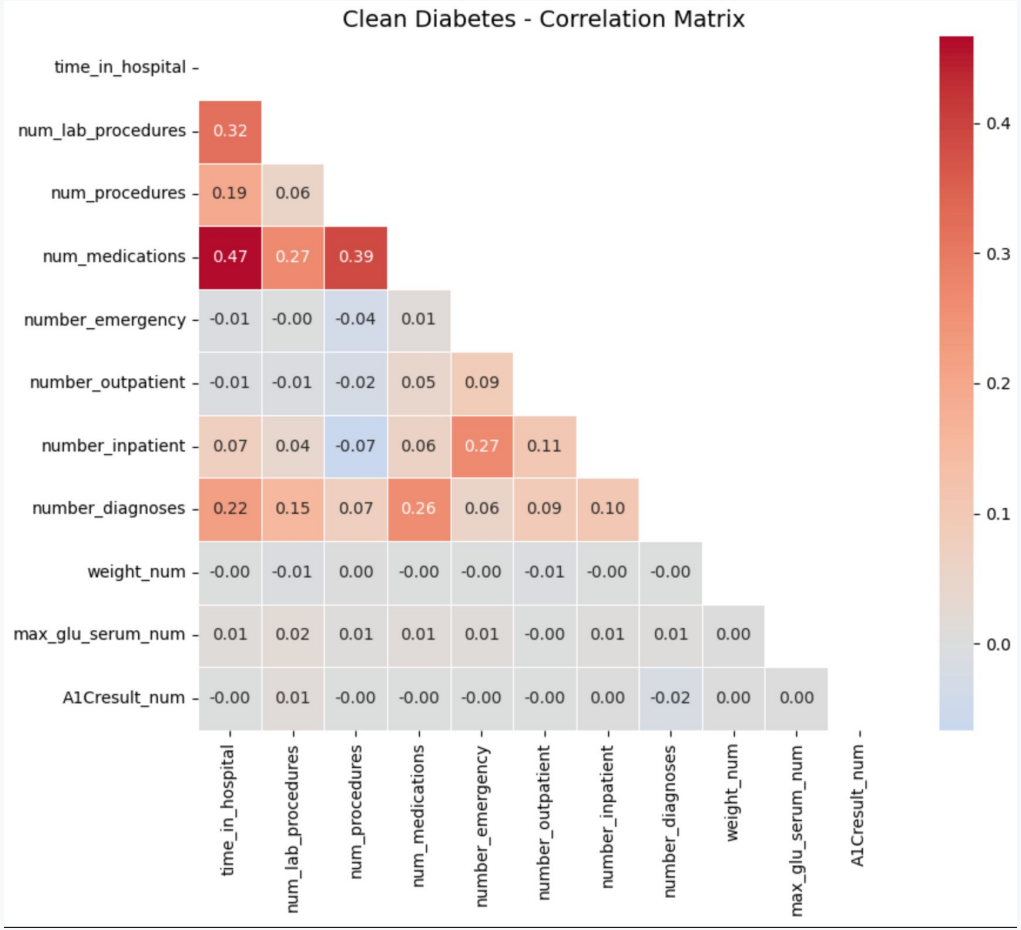
Hyperparameter Tuning

- ✓ Best 3 models optimized
- ✓ Recall prioritized
- ✓ Overfitting prevented
- ✓ Final **models** selected

Least data exclusion - Cleaned data - Clinically relevant signals - Tested multiple approaches - Fine-tuned the winner

>> To maximize ability to catch highest-risk patients <<

02 Correlation Matrix



03 Model Comparison

Out [125...

	Model	ROC-AUC	Accuracy	Recall	Precision	Specificity
0	Logistic Regression	0.665520	0.879332	0.058924	0.364912	0.986593
1	KNN	0.580742	0.216246	0.926912	0.121446	0.123333
2	SGDClassifier	0.596633	0.448739	0.708215	0.136612	0.414815
3	Decision Tree	0.581884	0.785064	0.201700	0.159785	0.861333
4	Random Forest	0.566407	0.257190	0.882720	0.122774	0.175407
5	Linear SVC	0.668318	0.880904	0.047592	0.380090	0.989852
6	Gradient Boosting	0.646835	0.865313	0.113314	0.289436	0.963630
7	XGBoost	0.640255	0.865509	0.115014	0.292507	0.963630
8	CatBoost	0.652887	0.866295	0.116147	0.298834	0.964370
9	Decision Tree (HP Tuned)	0.511469	0.286734	0.789802	0.117035	0.220963
10	Random Forest (HP Tuned)	0.564490	0.115624	1.000000	0.115624	0.000000
11	XGBoost (HP Tuned)	0.617336	0.846315	0.152408	0.240393	0.937037

03 Evidence of Value

Accuracy

86%CB

Correct overall predictions

Recall

~88%RF

At-risk patients correctly flagged

Specificity

96%CB

Flags that are truly high risk

ROC-AUC

.X

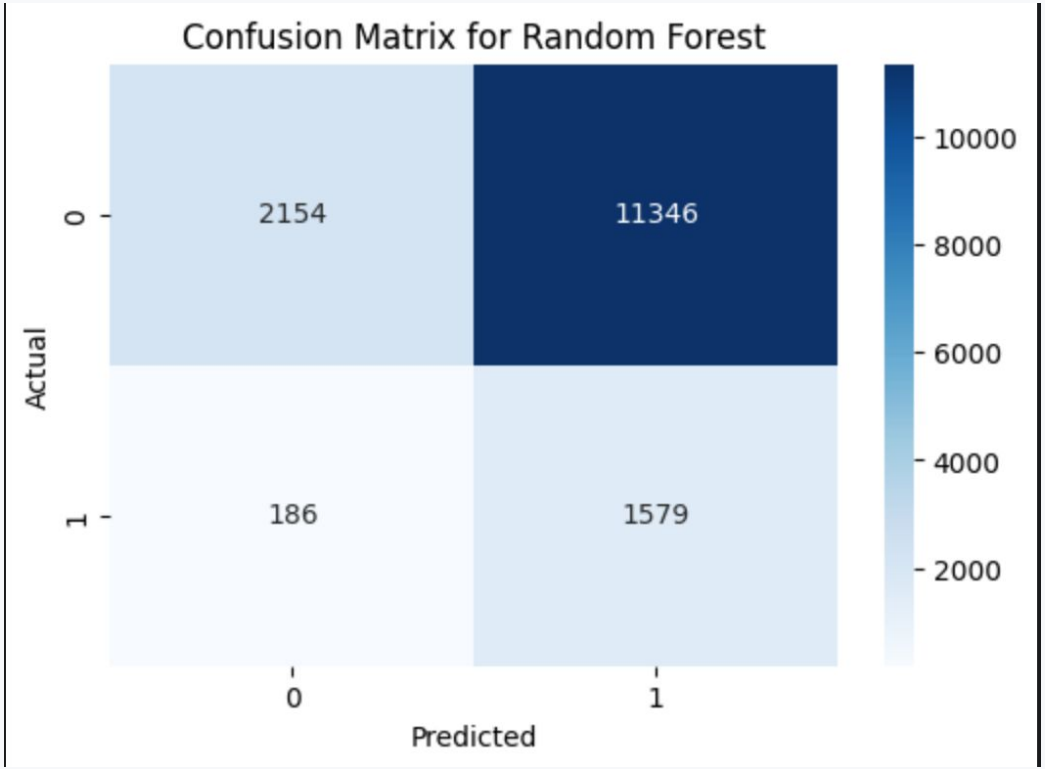
Discrimination ability (0-1 scale)

Why We Prioritize Recall

In a clinical context, missing a high-risk patient is more costly than a false alarm.

92% recall means for every 100 at-risk patients, we correctly flag 92 - giving care teams a meaningful head start.

03 Evidence of Value



Model trained and validated on UCI 130-hospital diabetes dataset (n ≈ 101,766 encounters).
Local fine-tuning on hospital EHR data will improve AUC to an estimated 0.75 – 0.85.

04 Model Demo

Streamlit App

Individual patient alert system

04 How the Model was built

1	2	3	4	5
Data Foundation	**Clinical Feature Selection**	**Rigorous Testing**	**Explainability Built In**	**Fairness Validation**
Started with 101,766 real diabetes discharge records with 50 features from 130 US hospitals. Cleaned for missing values etc, and inconsistent coding thus ensuring the model learns from reliable, unbiased (?) data.	Identified the factors clinicians already know matter: HbA1c levels, prior admission count, medication changes, discharge destination, and social risk ?. The model speaks the same language as the care team.	Multiple model types compared. Final model (XGBoost) was fine-tuned to maximize **RECALL** (measures the ability of a model to correctly identify all relevant instances, specifically focusing on the proportion of actual positives that are captured) and **SPECIFICITY** (catching the most high-risk patients - while minimizing unnecessary alerts).	Every risk score comes with a plain-language explanation: the top 3 reasons why this patient is flagged. No black box, clinicians can see, question, and override the reasoning.	The model can be tested separately across racial, age, and payer subgroups. It is designed to perform equitably, no demographic group is over-alerted or under-protected. ??

ML Model is like a very experienced colleague who has reviewed 101,766 similar cases and gives you a second opinion at discharge , without adding to workload.

04 How the Model works: predictive features

These 8 factors carry the most predictive weight [5] - all already documented in Epic:

#1 Prior inpatient admissions

Count in last 12 months - strongest single predictor



#2 HbA1c at discharge

Values >9% significantly increase risk



#3 Discharge disposition

Home without home health is highest risk



#4 Insulin regimen change

Change during admission signals instability



#5 Payer type

Medicaid patients face systemic access barriers



#6 Length of stay

Longer stays correlate with severity



#7 Medication count

Polypharmacy increases complexity at home



#8 Area Deprivation Index

ZIP-code social risk (ADI score)



04 How the Model was built

FILTERING

Exclusion Criteria

- indexing info
- payer code
- medical specialities
- medication change
- individual meds (except Insulin)

OPTIMIZATION

Feature Engr.

- 23 features
- 115 columns
- 101,766 rows
- cleaned
- imputed
- mapped
- one hot encoded

PATHWAY

Methods

- EDA
- skewness
- correlations (heatmap)
- imbalance - 5 RO, RU, SMOTE SMOTE-TOMEK, **SMOTE-ENN**

SPLIT & TUNING

Training Suite

- 71,236 Train
- 15,265 Validation
- 15,265 Test
- Stdscalar
- Hypertuning

RF - XGBoost - Decision Tree

ALGORITHMS

ML Classifiers

Log Reg	Random RF
KNN	G. Boosting
SGDC	XGBoost
Decision Tree	CatBoost
Linear SVC	

AUC - .Accuracy - **Recall**
Precision -.**Specificity**
AUC-ROC Curve
F1 Score - Sensitivity - Log Loss

Final Outcomes

Metric	Recall	Specificity
COMPARISON	TP / (TP = FN)	TN / (TN + FP)

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05 Workflow Integration: What Clinicians See

Admission	During stay	Discharge order	Post-discharge
Background data pull	**Risk factors update**	**Risk alert fires**	**Patient notified**
Epic automatically pulls labs, meds, history. No clinician action required.	Model refreshes with new labs and medication changes. Score updates silently.	If score ≥ 40: BPA appears in discharge workflow with top-3 reasons and one-click care plan.	MyChart message sent with self-care guidance and prompt to schedule 7-day follow-up.
Automatic	Automatic	Physician sees alert	Patient receives

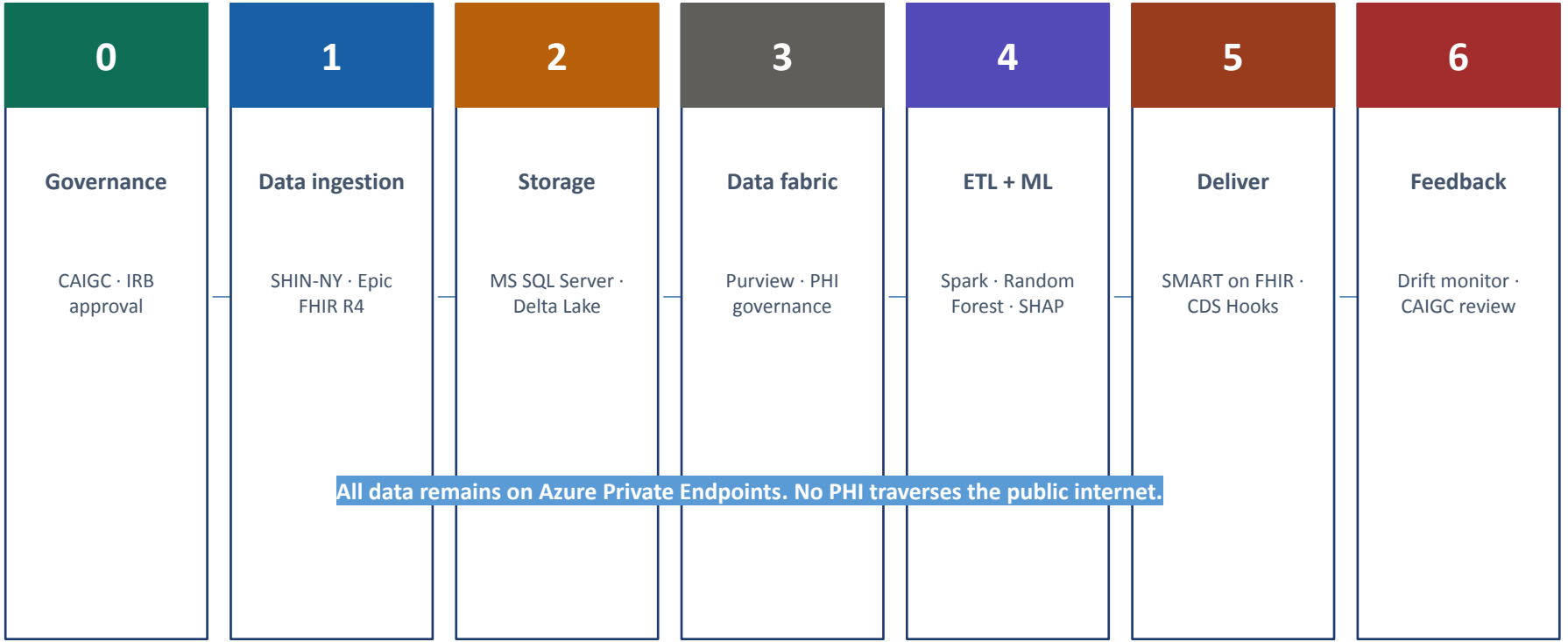
Zero new logins. Zero new screens. The tool surfaces in Epic where the work already happens.

05 Clinical Workflow Integration

Designed to fit into existing workflows - zero new systems required.

Admission	Patient data begins auto-populating from EHR - no extra charting required
During Stay	Model continuously updates risk score as labs, meds, and diagnoses are added
Pre-Discharge	Risk flag appears in discharge planning view - HIGH / MODERATE / LOW displayed
Care Huddle	Team reviews flag alongside clinical judgment - care plan adjusted accordingly
Follow-Up Set	High-risk patients auto-scheduled for 7-day call and 14-day clinic appointment

06 Architecture: 7-Phase Implementation



NYS Compliance Coverage:

- HIPAA
- NY SHIELD Act
- HITECH
- 42 CFR Pt 2
- NY DOH AI Reporting
- Epic App Orchard

06 Data Sources - What Feeds the Model

EHR

Epic EHR

FHIR R4 - labs, meds, diagnoses, discharge notes

HIE

SHIN-NY HIE

Out-of-hospital encounters at other NY facilities

ADT

ADT Events

Real-time admission/discharge triggers (HL7 v2)

CDC

BRFSS Survey

CDC county-level behavioral risk data (smoking, obesity)

CMS

CMS PUF / SPARCS

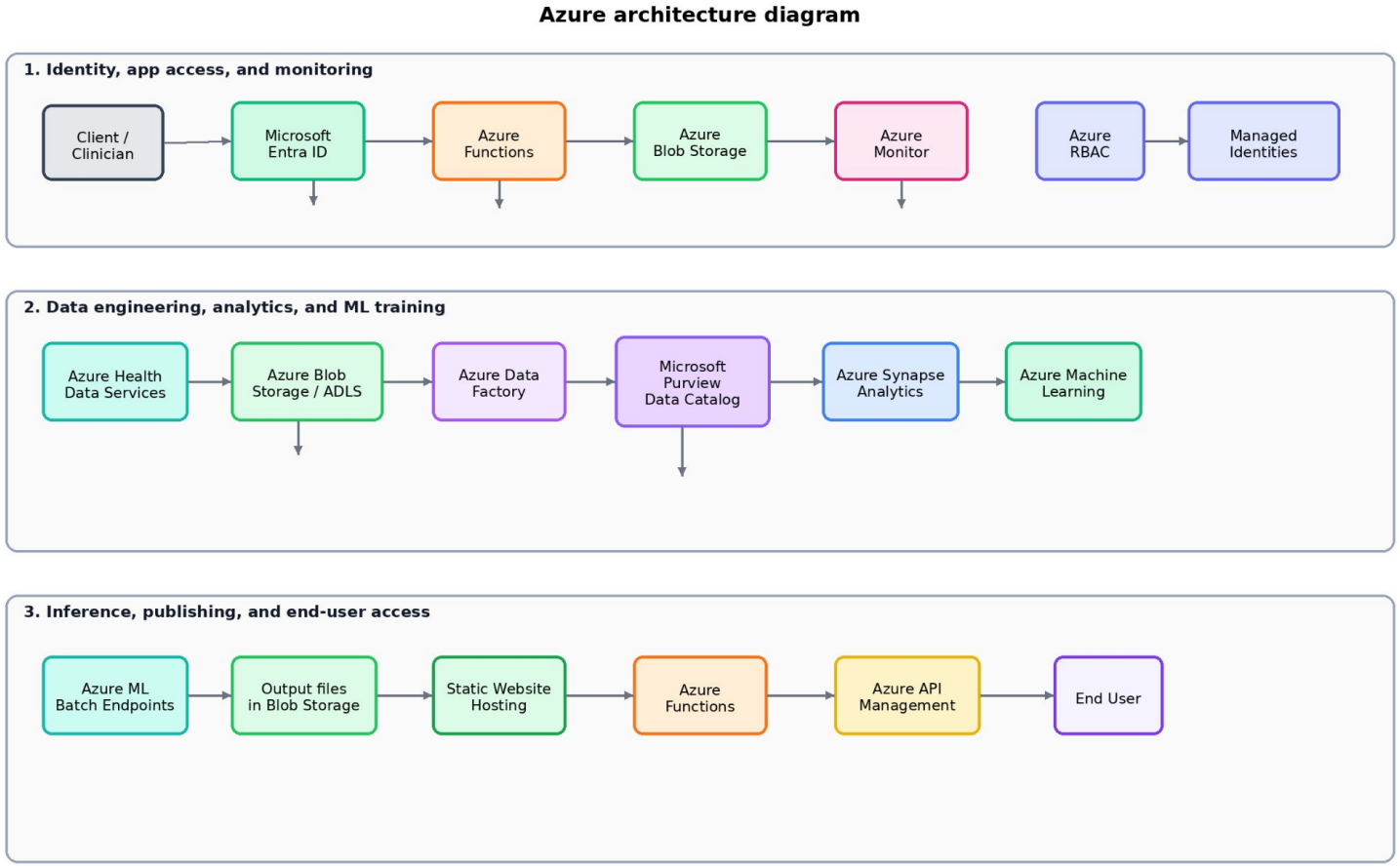
NYS statewide readmission benchmarks for calibration

WRB

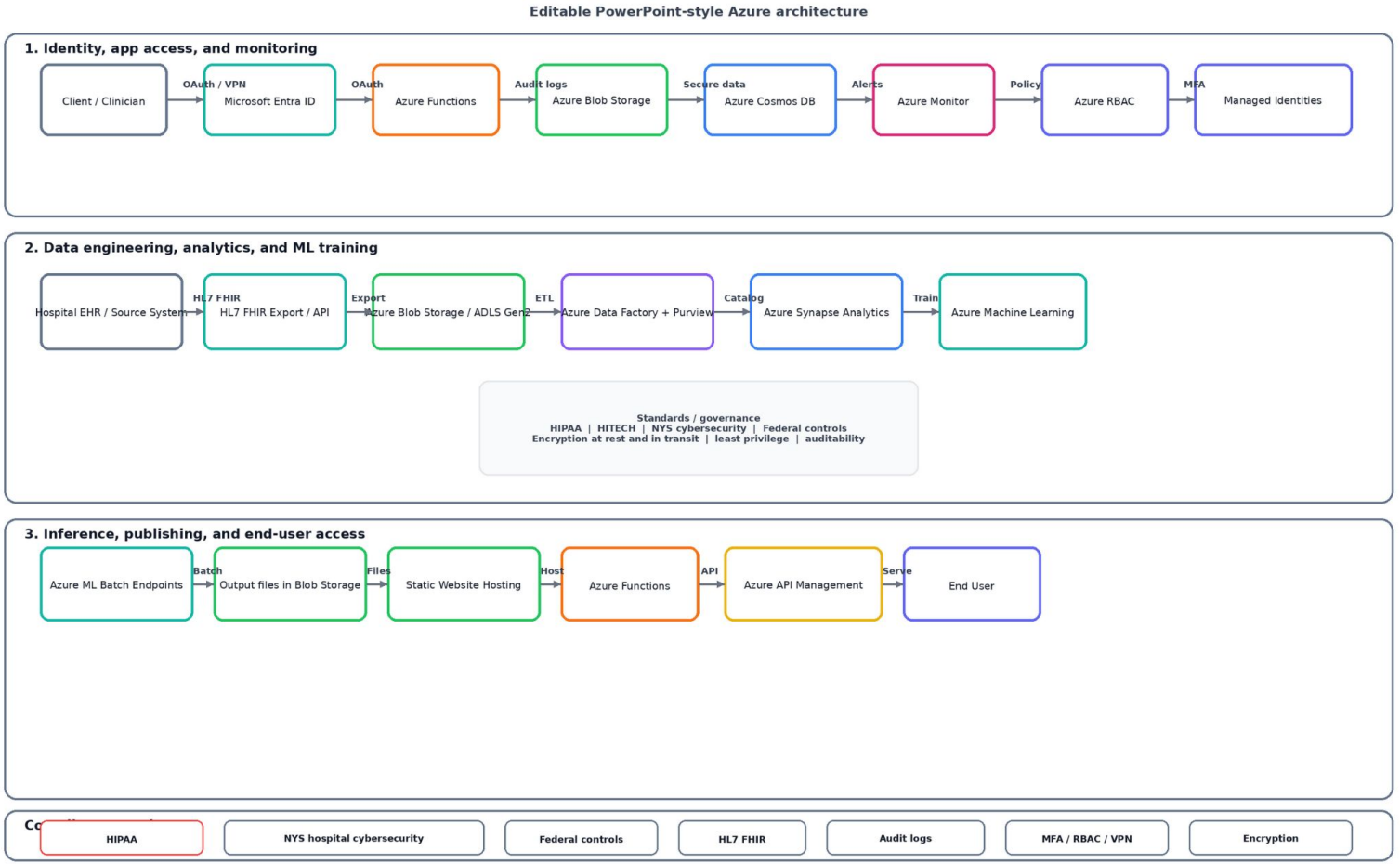
Patient Wearables

HealthKit glucose and activity data (opt-in)

06 Architecture



06 Architecture - with data stds



06 Pipelines

1

Source Systems

EHR, claims, labs, meds, diagnoses, encounters, and discharge data come in as FHIR resources or FHIR-exported data. > FHIR Ingestion

Governance Layer

Validate data quality, enforce access rules, thresholds, track lineage, manage consent, and approve which features are allowed into modeling.

5

Visualization Layer

Present risk scores, explanations, and trends in a SMART on FHIR app embedded inside the EHR workflow.

2

ETL Layer

Extract the FHIR JSON, transform it into tabular patient-encounter rows, clean missing values, encode categorical fields, and create readmission labels/features.

3

ML Model

Feature store & training set. Train and validate models on the curated dataset, then score each encounter with a predicted readmission probability. Thresholding.

4

Decision layer

Apply a tuned threshold to turn probabilities into alerts, to balance catching high-risk patients with reducing false alerts.

07 Governance, Safety & Human Oversight

Every safeguard in this model was designed with clinical reality in mind:

Advisory only

The model never orders, restricts, or overrides. It presents information. Every clinical decision remains with the physician.

Explainable by design

Every alert includes the top 3 clinical reasons in plain language. No black box: clinicians can review, challenge, and dismiss.

Clinician feedback loop

Every dismissed alert is logged with a reason. The governance committee reviews this data at 30/60/90-day checkpoints to recalibrate thresholds.

Fairness monitoring

AUC performance is tracked separately by race, payer, and age group. If any group shows more than a 5-point gap, the model is flagged for review.

Drift detection

The model's accuracy is checked monthly against actual readmission outcomes. If performance drops below threshold, automated retraining fires.

IRB & CAIGC oversight

The Clinical AI Governance Committee (CMO, CNO, CISO, IRB) approves all model versions before production deployment. No deployment without sign-off.

This model complies with FDA CDSS advisory-only guidance (21st Century Cures Act) and does not require 510(k) clearance as currently designed.

07 Safety, Governance & Responsible AI

Human-in-the-Loop

Every model flag is a recommendation only. Clinicians retain full authority over all patient care decisions.

Model Transparency

We can explain which factors drove any individual risk score — no black box decision-making.

Bias Monitoring

Ongoing monitoring across age, gender, and payer groups ensures equitable performance for all patients.

Data Privacy & HIPAA

All data is de-identified during development. Production model operates within existing HIPAA-compliant EHR infrastructure.

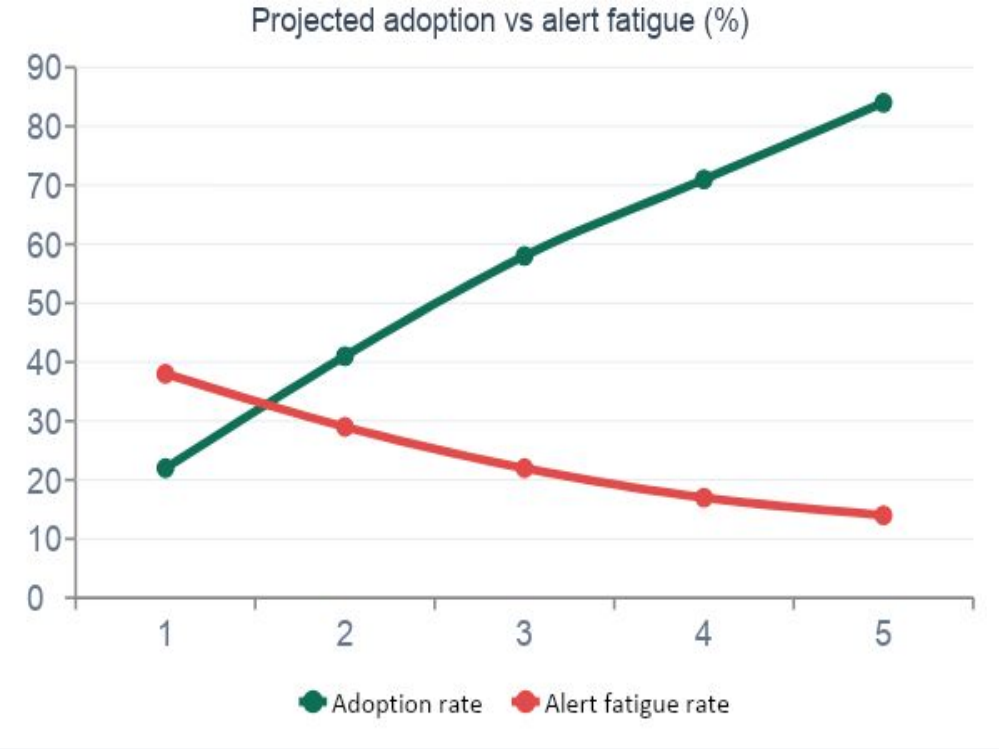
Continuous Validation

Model performance will be re-evaluated every 6 months against real-world outcomes and retrained as needed.

Known Limitations

Model trained on historical data; accuracy may vary across subpopulations not well-represented in the dataset/data gaps.

07 Change Management: Making Adoption Succeed



Our four commitments to your team:

Role-specific training

Separate sessions for physicians, nurses, case managers, and pharmacists. Each sees only what is relevant to their workflow.

Threshold control

The CAIGC sets and adjusts the alert threshold. If alert volume is too high, the committee lowers it — no ML team involvement needed.

30/60/90 review cadence

Structured governance reviews at 30, 60, and 90 days with Power BI dashboards showing alert fatigue, equity, and outcome data.

Patient communication

Every patient receives a MyChart message explaining the tool exists and is advisory-only, with an opt-out option.

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Timeline, investment ask & proposal

08 Implementation Timeline

Months 1–2

Foundation & governance

- CAIGC formation and approval
- IRB submission and review
- Azure infrastructure setup
- Staff communication plan

Months 3–4

Data & model integration

- SHIN-NY gateway connection
- Epic FHIR R4 API integration
- MS SQL Server + Delta Lake setup
- Feature store and ML pipeline validation

Month 5

Pilot launch

- SMART on FHIR app deployment
- CDS Hooks BPA activation (endocrinology unit only)
- Role-specific training complete
- 30-day review cadence begins

Month 6+

Full deployment & monitor

- Hospital-wide rollout
- Power BI + Tableau dashboards live
- Automated drift monitoring active
- 60/90-day CAIGC reviews

08 Investment & Expected Return

Estimated annual operating cost [6]

Component	Annual cost
Azure cloud infrastructure (ADLS, SQL MI, AML)	$85,000–$120,000
Databricks platform license	$40,000–$60,000
Microsoft Purview + Sentinel	$18,000–$28,000
Epic SMART app + App Orchard fee	$15,000–$25,000
Staff training + change management	$20,000 (Year 1)
TOTAL ESTIMATED	$178,000–$233,000/yr

Financial estimates based on NYS CMS penalty data, local admission volume, and published intervention literature.

Expected return

$2.1M
CMS penalty exposure avoided
(at current readmission rate)

$550K
Estimated direct cost savings from prevented readmissions

8.3×
Projected ROI in Year 1 based on mid-range estimates

Anticipated Questions

Q: Does this require an Epic upgrade? — IT staff

No. The SMART app deploys via Epic App Orchard framework. CDS Hooks uses Epic's existing BPA infrastructure. No Epic version change required.

Q: Who owns the alert threshold decision? — Clinicians

The Clinical AI Governance Committee, not the ML/tech team. The committee sets, adjusts, and is accountable for the threshold policy.

Q: What if a physician disagrees with the score? — Clinicians

They dismiss it with a reason code. All dismissals are reviewed. If patterns emerge, the threshold or model is recalibrated.

Q: How is patient data protected? — Compliance

All data stays on Azure Private Endpoints inside the hospital's VNet. No PHI traverses the public internet. HIPAA, NY SHIELD, and 42 CFR Pt 2 compliant.

Q: What about AI bias against minority patients? — Clinical equity

Fairness audits run at every release. If any demographic subgroup shows more than a 5-point AUC gap, the model is blocked from deployment until resolved.

Q: Is this FDA-regulated? — Legal / compliance

As an advisory-only CDSS where the physician can independently review the basis, this currently falls under the 21st Century Cures Act advisory CDSS exemption from FDA 510(k).

Approve a 6-month 2-unit pilot beginning Q3 2026

If the pilot demonstrates an alert fatigue rate below 20% and a sensitivity above 65%, we recommend hospital-wide rollout in Month 7.

1	Convene the CAIGC and appoint the IRB lead
2	Approve pilot budget ($178K-$233K Year 1)
3	Authorize Epic App Orchard submission